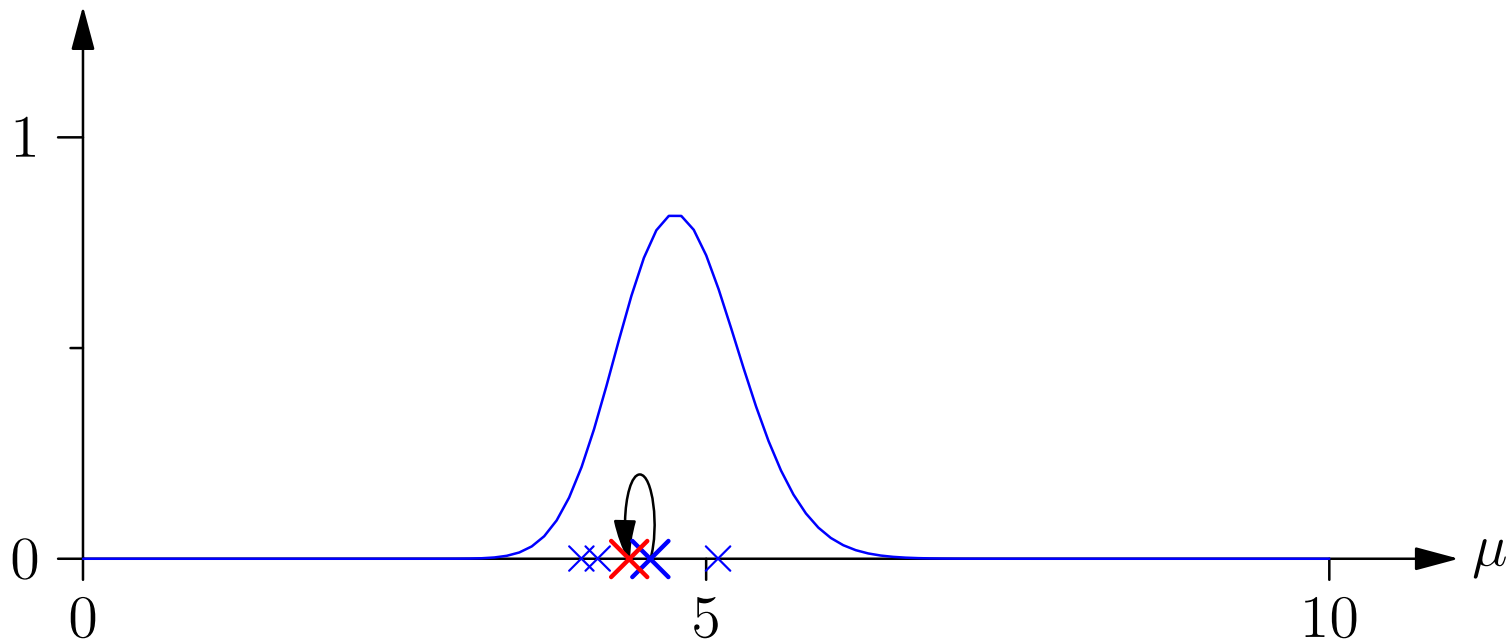


Mathematics of Machine Learning

MCMC

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2, 3, 11\}$$

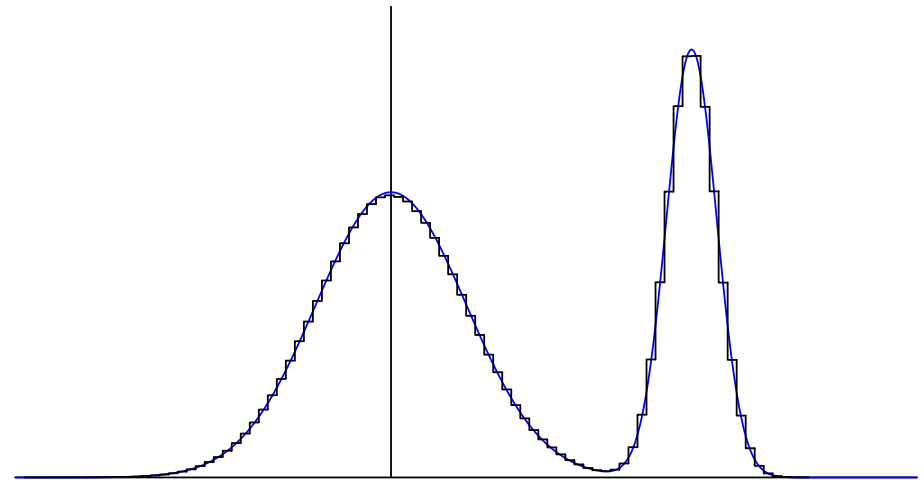


Monte Carlo methods, MCMC, Variational Methods

Outline

1. **Sampling**
2. Random Number Generation
3. MCMC
4. Quiz

$T = 10000000$, acceptance rate = 0.897



Bayesian Inference Gets Hard

- We saw that in some cases if we had a simple likelihood (normal, binomial, Poisson, multinomial) you can choose a conjugate prior (gamma-normal/Wishart, beta, gamma, Dirichlet) so that the posterior has the same form as the prior■
- Very often we are working with more complex models where no conjugate prior exists■
- The posterior is not described by a known distribution■
- We have to work a lot harder—particularly with multivariate distributions■

Bayesian Inference

- Recall our problem is that we are given some data $\mathcal{D}$ ■
- Our posterior is given by

$$\mathbb{P}(\boldsymbol{\theta}|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\boldsymbol{\theta})\mathbb{P}(\boldsymbol{\theta})}{\mathbb{P}(\mathcal{D})} \quad \text{or} \quad f(\boldsymbol{\theta}|\mathcal{D}) = \frac{f(\mathcal{D}|\boldsymbol{\theta})f(\boldsymbol{\theta})}{f(\mathcal{D})}$$

- Where $\boldsymbol{\theta}$ are the parameters we are trying to infer■
- But our likelihood (and/or prior) might be quite complicated■
- Typically we don't have a closed form representation for our posterior distribution■

Complex Likelihoods

- In the classical problems with parameters, θ , that are well known statistics such as the mean and variance■
- But suppose we are trying to distinguish between different models of cosmological inflation (physics models of the early Universe)■
- Our data is the Cosmic Microwave Background measure by a satellite telescope■
- Our parameter, θ , might represent unknown parameters of the early universe (width of surface of scattering, number of e-foldings, primordial perturbations, amount of cool dark matter)■
- Here the likelihood $\mathbb{P}(\text{CMB} \mid \theta)$ is far from a well known probability distribution■

Histograms, Samples and Means

- We could represent our posterior as a histogram, although for multivariate distributions (i.e. when we are modelling more than one variable) a histogram can be unwieldy■
- A sample from the posterior distribution is often sufficient e.g. in our topic models (LDA) a typical set of topics is what we are after■
- However, when samples vary a lot, often the most useful quantities are expectation, e.g.

$$\mathbb{E}[\Theta]$$

$$\mathbb{E}[\Theta_i \Theta_j] - \mathbb{E}[\Theta_i] \mathbb{E}[\Theta_j]$$

$$\mathbb{E}[\Theta_i^2] - \mathbb{E}[\Theta_i]^2$$

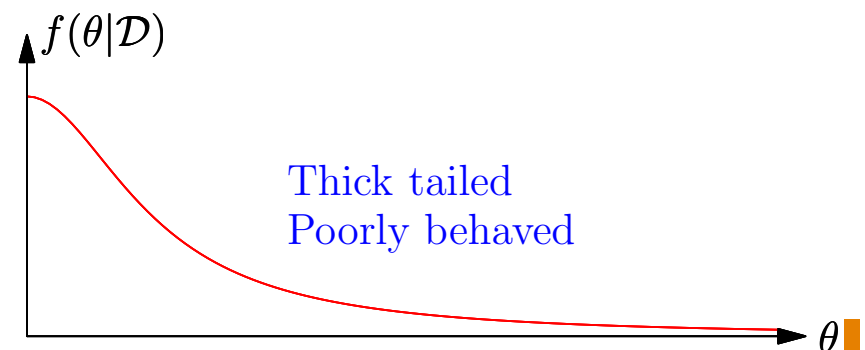
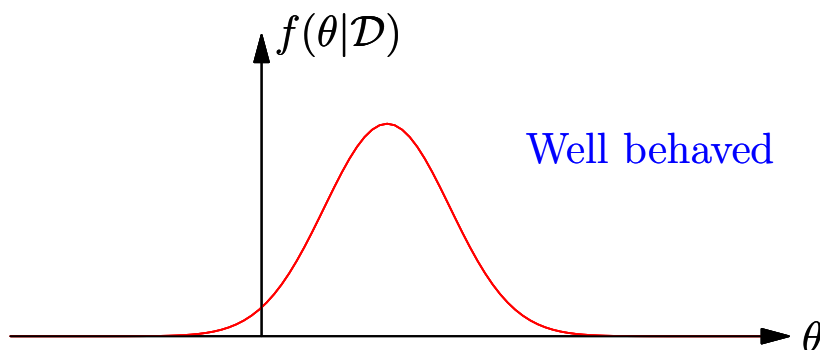
$$\mathbb{E}[\Theta \Theta^T] - \mathbb{E}[\Theta] \mathbb{E}[\Theta]^T \blacksquare$$

Sample Estimation

- If we can draw independent samples (aka **deviates** or **variates**), Θ_i , from our posterior distribution then we can obtain an estimate of our expectation

$$\mathbb{E}[g(\Theta)] \approx \frac{1}{n} \sum_{i=1}^n g(\Theta_i)$$

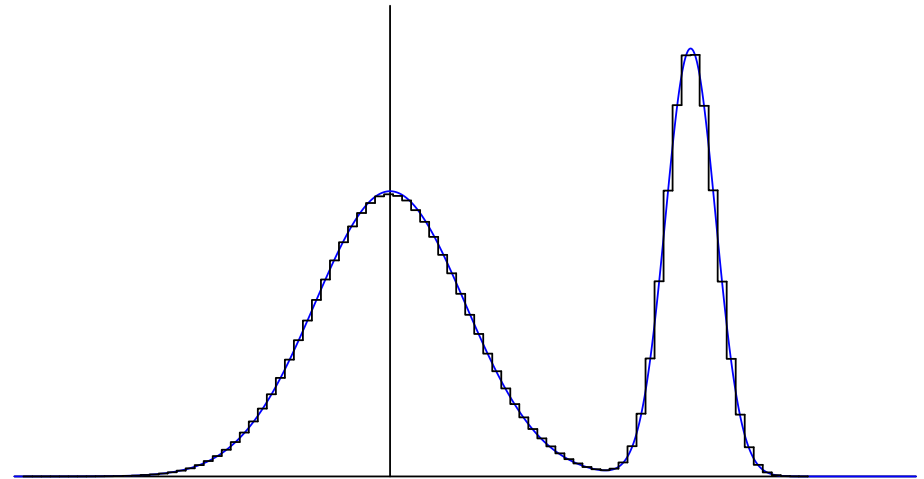
- Provided our posterior distribution is well behaved the relative error in our estimate will drop off as $1/\sqrt{n}$



Outline

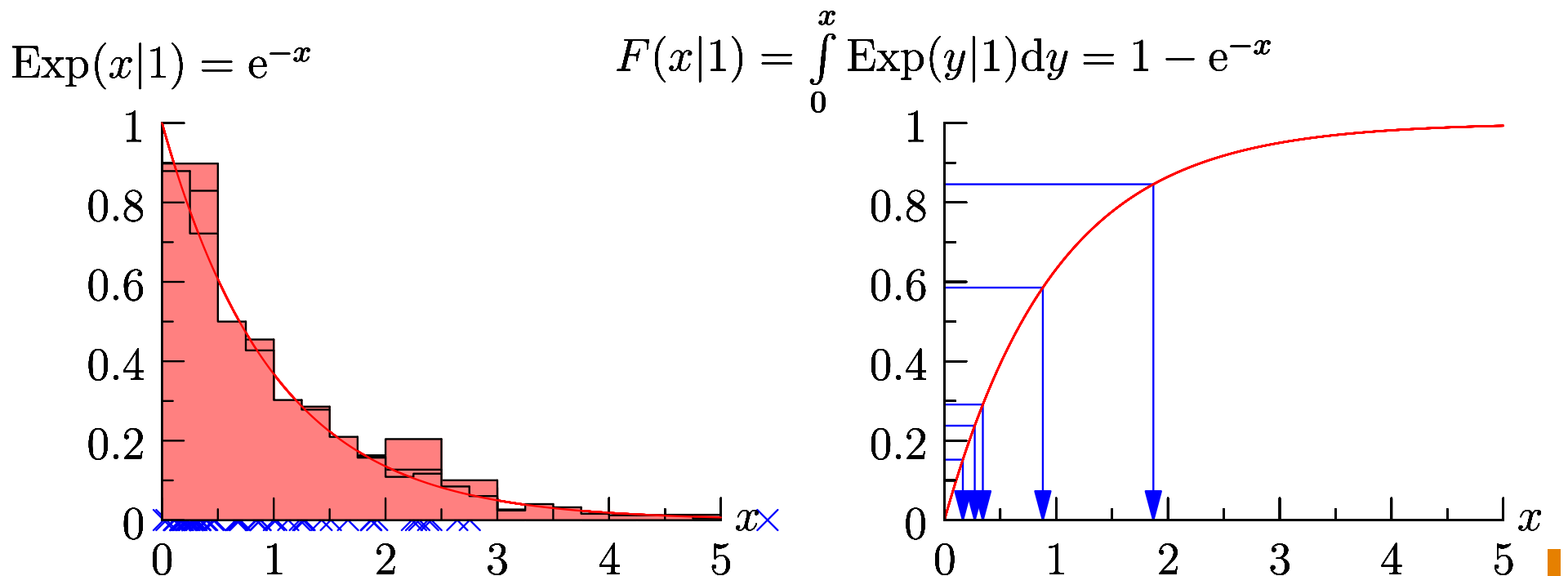
1. Sampling
2. **Random Number Generation**
3. MCMC
4. Quiz

$T = 10000000$, acceptance rate = 0.897



Drawing Random Samples

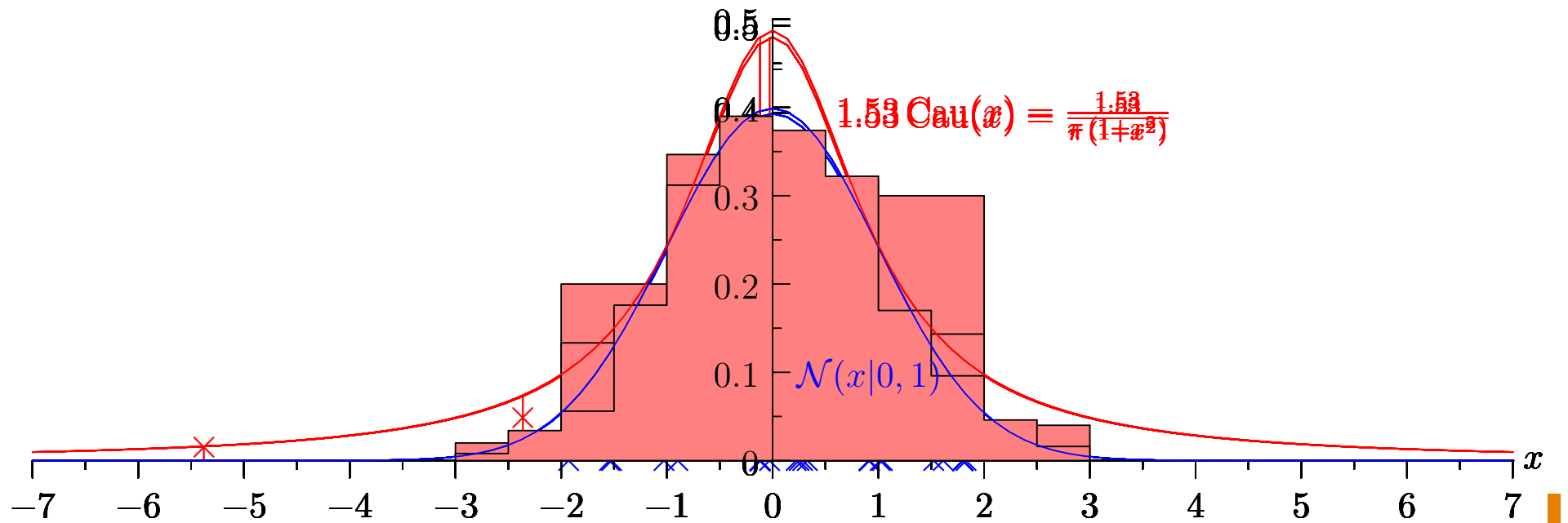
- Drawing (pseudo) random variables from a distribution is known as **Monte Carlo**
- For some very simple distributions we can use the **transformation methods** to transform a uniform distribution



Rejection Method

- The transformation method only works when we can easily compute the inverse *cumulative distribution function* (CDF)■
- A more general technique is the **rejection method** where we generate deviates from $g_Y(y)$ such that $cg_Y(x) \geq f_X(x)$ ■
- To draw deviates from $f_X(x)$ we draw a deviate $Y \sim g_Y$ and then accept the deviate with probability $f_X(Y)/(cg_Y(Y))$ ■
- The expected rejection rate is $c - 1$ ■
- Need to choose a good distribution $g_Y(y)$ ■

Drawing Normal Deviates



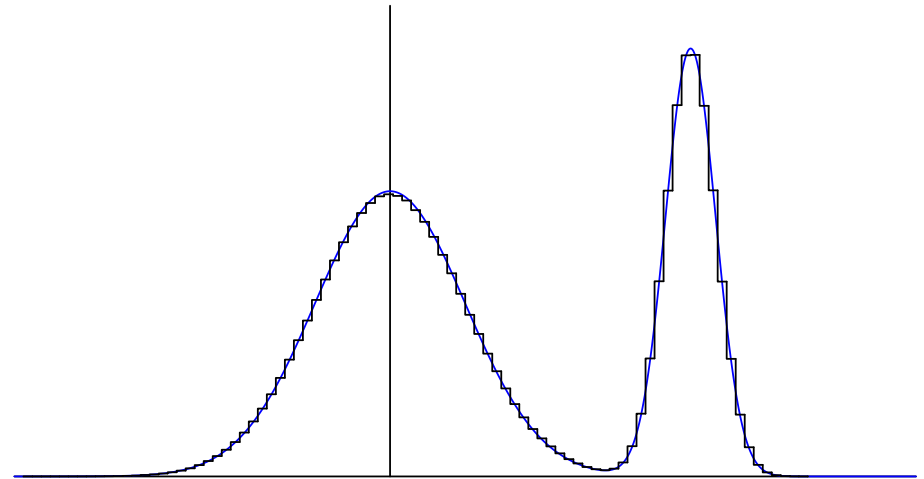
Problems with Rejection

- The rejection method is very general and often the method of choice (although for normal deviates there is a clever transformation method which is faster)■
- However, for complicated probability distributions it can be difficult to find a good proposal distribution $g_Y(y)$ ■
- This is particular true for multivariate distributions■
- If the proposal distribution is poor c might be very high and the number of rejections is stupidly high■

Outline

1. Sampling
2. Random Number Generation
3. **MCMC**
4. Quiz

$T = 10000000$, acceptance rate = 0.897



Detailed Balance

- Suppose we have a set of states $\mathcal{S}$ and want to draw sample from a probability distribution $\pi = (\pi_i | i \in \mathcal{S})$ ■
- We invent a dynamical system with a transition probability M_{ij} from state j to state i such that

$$M_{ij}\pi_j = M_{ji}\pi_i$$

- This is known as **detailed balance**■
- Summing both sides over j

$$\sum_j M_{ij}\pi_j = \sum_j M_{ji}\pi_i \quad \text{■} = \pi_i \quad \text{■} \quad \mathbf{M}\pi = \pi \quad \text{■}$$

Convergence of MCMC

- Suppose we start from a state $\mathbf{x}(0) = \sum_i c_i \mathbf{v}^{(i)}$ where the $\mathbf{v}^{(i)}$'s are an eigenvectors of the transition matrix $\mathbf{M}$ with eigenvalues λ_i
- If I apply $\mathbf{M}$ many times then

$$\mathbf{x}(t) = \mathbf{M}^t \mathbf{x}(0) = \mathbf{M}^t \sum_i c_i \mathbf{v}^{(i)} = \sum_i \lambda_i^t c_i \mathbf{v}^{(i)}$$

- And $\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{v}^*$ where $\mathbf{v}^*$ is the eigenvector with the maximum eigenvalue
- Now $\|\mathbf{M}\mathbf{v}\|_1 \leq \|\mathbf{M}\|_1 \|\mathbf{v}\|_1 = \|\mathbf{v}\|_1$ so the maximum eigenvalue is 1 with eigenvector $\boldsymbol{\pi}$ ($\mathbf{M}$ is known as a **stochastic matrix**)

Metropolis Algorithm

- A very easy way to achieve detailed balance is starting from state j choose a “neighbouring” state, i with equal probability■
- We accept the move if either
 - ★ $\pi_i > \pi_j$ or
 - ★ we make the move with a probability π_i/π_j ■
- If $\pi_i > \pi_j$ then $M_{ij} = 1$ and $M_{ji} = \pi_j/\pi_i$. Thus

$$M_{ij}\pi_j = \pi_j \quad M_{ji}\pi_i = \frac{\pi_j}{\pi_i}\pi_i = \pi_j$$

- Note that we require the state i to have the same number of neighbours as state j so that detailed balance is satisfied■

Continuous Variables

- If we are working with continuous variables θ then the equation for detailed balance for the transition probability $W(\theta \rightarrow \theta')$ is

$$W(\theta \rightarrow \theta')\pi(\theta) = W(\theta' \rightarrow \theta)\pi(\theta') \blacksquare$$

- where $\pi(\theta)$ is the probability distribution we wish to sample from $\blacksquare$
- The update rule is to choose a nearby value θ' , compute $r = \pi(\theta')/\pi(\theta)$ and accept the update with probability $\min(1, r)$ $\blacksquare$
- We require that the probability of choosing θ from θ' is the same as the reverse $\blacksquare$

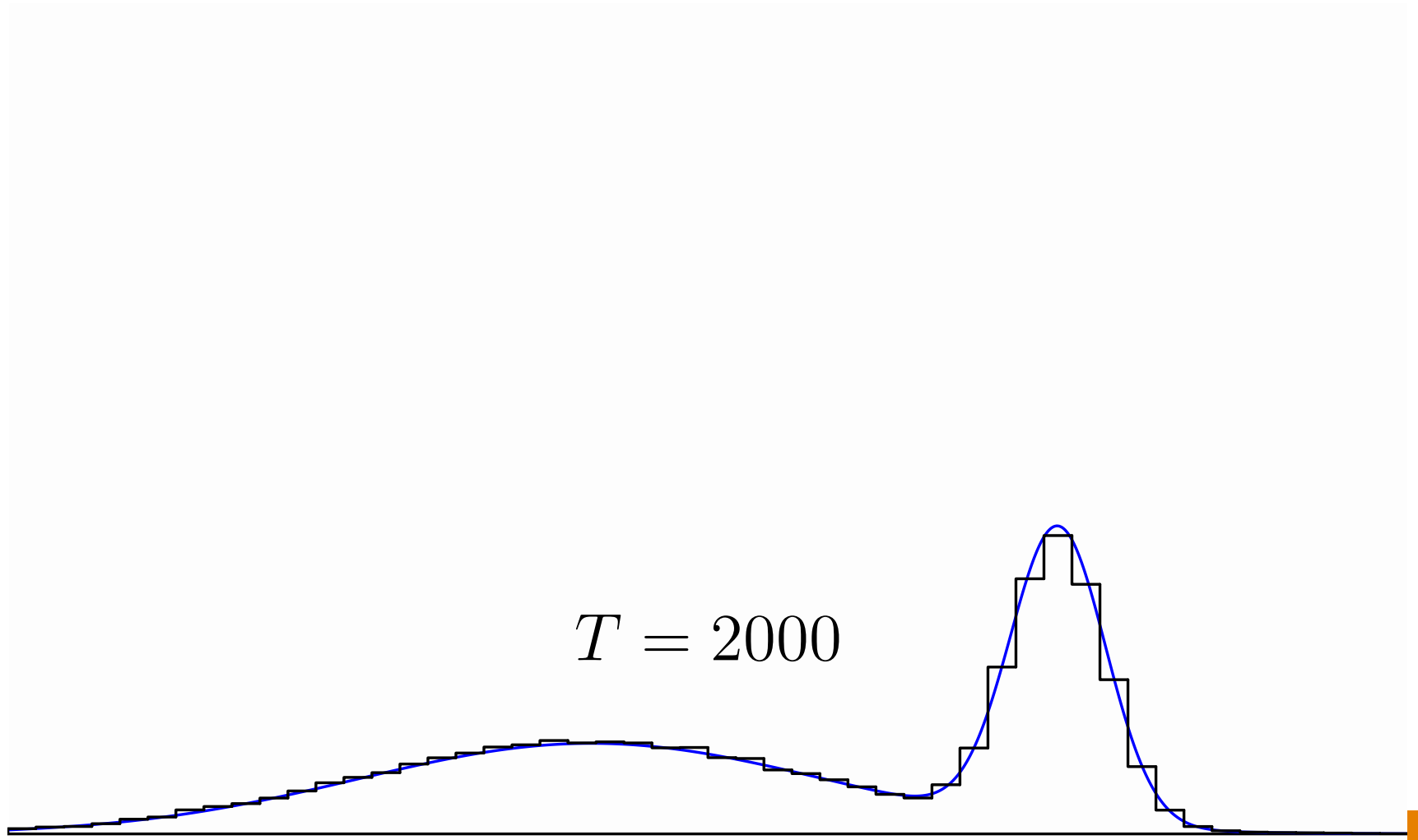
What Makes MCMC Nice

- Because we are free to choose where we move (and choose close by neighbours) $\pi(\boldsymbol{\theta}') \approx \pi(\boldsymbol{\theta})$ so that moves are not too infrequent■
- Also very importantly the updates depend only on the ratio $\pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta})$ ■
- We only need to know our probabilities up to a multiplicative scaling factor■
- For sampling from the posterior we only need to know the likelihood and prior $\mathbb{P}(\mathcal{D}|\boldsymbol{\theta})\mathbb{P}(\boldsymbol{\theta})$ ■ (or $f(\mathcal{D}|\boldsymbol{\theta})f(\boldsymbol{\theta})$)■
- We don't need to know $\mathbb{P}(\mathcal{D})$ which we generally don't know■

What Makes MCMC Nasty

- It can take a long time until our states occur with the probability π (i.e. we have forgotten our initial state)■
- We don't even know how long we have to wait■
- Even when we have reached this *equilibration time* each sample is correlated with the previous sample■
- To get a good approximation to the posterior expectation requires running for many times the equilibration time■
- Note, if we are just finding sample averages then we can use all samples after equilibrating even if they are not independent■

Burn-In



Proposals and Metropolis-Hastings

- We have some freedom in choosing a new proposal θ' from our current position θ —a good choice can increase the acceptance rate making the MCMC more efficient■
- We define the proposal distribution $p(\theta'|\theta)$ ■
- For the standard Metropolis algorithm to work we require $p(\theta'|\theta) = p(\theta|\theta')$ ■
- In some cases (e.g when $\theta_i \geq 0$) this can be hard to achieve■
- We can modify our update rule to accept a move with probability

$$\min\left(1, \frac{p(\theta|\theta') f(\mathcal{D}|\theta') f(\theta')}{p(\theta'|\theta) f(\mathcal{D}|\theta) f(\theta)}\right) \blacksquare$$

Traffic Rate

- Consider monitoring the flow of traffic where we have data

$$\mathcal{D} = (N_1, N_2, \dots, N_n)$$

where N_i is the number of car that past on day i ■

- We assume $N_i \sim \text{Poi}(\mu)$ and want to infer μ ■
- The Poisson distribution has a beta conjugate prior■
- We don't have any prior knowledge on μ so we use a non-informative prior $\text{Gam}(\mu|0,0) = 1/\mu$ ■
- Note that we can solve this problem exactly—however, lets compare with MCMC■

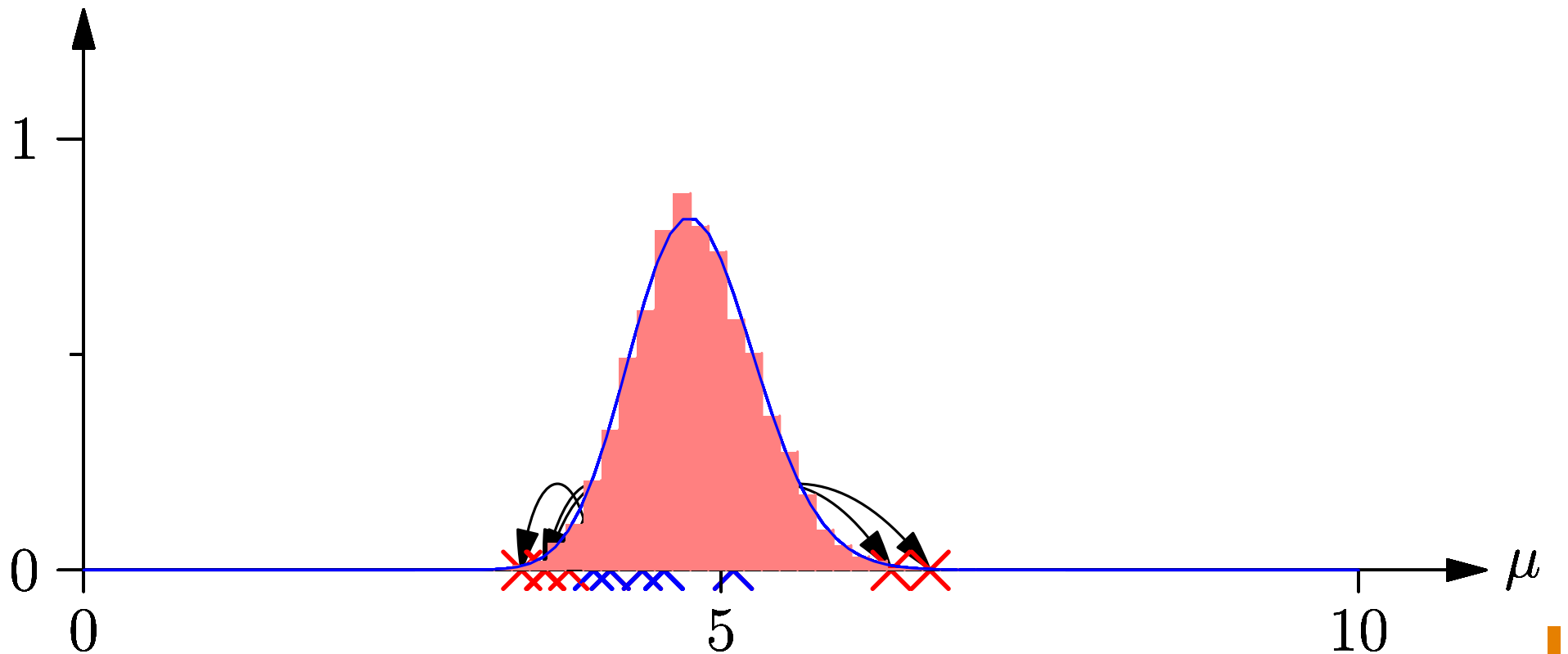
Proposal Distribution

- If we can choose our proposal distribution $p(\mu'|\mu)$ to be close to the posterior distribution then our acceptance rate would be close to 1■
- We choose $p(\mu'|\mu) = \text{Gam}(\mu'|\mu, \mu^2)$ which has $\mathbb{E}[\mu'] = \mu$ and variance 1■
- We update with probability $\min(1, r)$ where

$$\begin{aligned} r &= \frac{\text{Gam}(\mu|\mu'^2, \mu') \frac{1}{\mu'} \prod_{i=1}^n \text{Poi}(N_i|\mu')}{\text{Gam}(\mu'|\mu^2, \mu) \frac{1}{\mu} \prod_{i=1}^n \text{Poi}(N_i|\mu)} \blacksquare \\ &= \frac{\mu \text{Gam}(\mu|\mu'^2, \mu')}{\mu' \text{Gam}(\mu'|\mu^2, \mu)} e^{-n(\mu' - \mu) + \sum_{i=1}^n N_i \log\left(\frac{\mu'}{\mu}\right)} \blacksquare \end{aligned}$$

MCMC in Practice

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2, 3, 11\}$$



MCMC Details

- To compute correct histograms you need to count samples where no move is made multiple times■
- On modern computers its quite quick to compute millions of samples■
- The code is not very difficult to write (although care is need to get everything correct)■
- This can be used on complicated problems such as topic models (LDA) with thousands of parameters■
- The accuracy of MCMC is slow if it takes a long time to sample the posterior distribution■

The MCMC Industry

- MCMC provides a means to accurately sample from very complex models■
- There have been many advanced techniques developed to improve MCMC performance■
- E.g. hybrid MCMC simulates a dynamics to find good proposals with similar probability far from the starting point■
- Often it seems that MCMC is complicated because there are so many optimisations, but often simple implementations are sufficient■

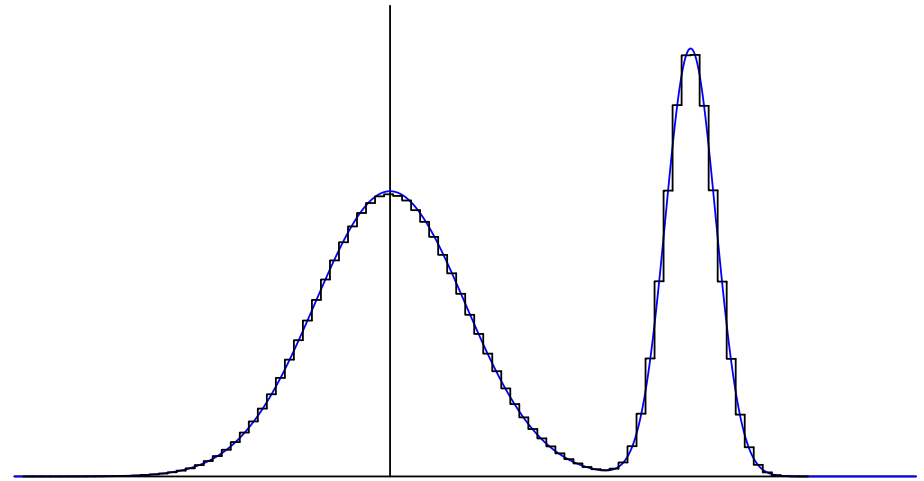
Conclusions

- As soon as we use complex models we are no longer able to compute the posterior in closed form■
- Monte Carlo techniques and particularly MCMC are a very general method for computing samples from the posterior■
- These techniques have been highly developed, but very frequently even simple implementations are sufficient to do good inference■
- MCMC is very easy to use, although it can be slow■
- There exist other methods such as variational approximations that can be much quicker, but are approximations■

Outline

1. Sampling
2. Random Number Generation
3. MCMC
4. Quiz

$T = 10000000$, acceptance rate = 0.897



Question 1

- How does the transformation method work to generate random deviates?
 1. You transform from one distribution to another
 2. You generate a uniformly distributed random number and apply the inverse probability distribution function to it
 3. You generate a uniformly distributed random number and apply the inverse cumulative distribution function to it
 4. You generate a uniformly distributed random number and apply the probability distribution function to it■
- Answer: 3. You generate a uniformly distributed random number and apply the inverse cumulative distribution function to it■

Question 2

- How does the rejection method work to generate random deviates?
 1. You generate from a nearby distribution and apply a transformation to the actual distribution
 2. You generate a random deviate, Y with a distribution g_Y such that $\forall x, cg_Y(x) \geq f_X(x)$ and then choose a uniform deviate U and accept Y if $Ucg_Y(Y) \leq f_X(Y)$
 3. You generate a random deviate, Y with a distribution g_Y such that $\forall x, cg_Y(x) \leq f_X(x)$ and then choose a uniform deviate U and accept Y if $Ucg_Y(Y) \geq f_X(Y)$ ■
- Answer: 2. You generate a random deviate, Y with a distribution g_Y such that $cg_Y(x) \geq f_X(x)$ and then choose a uniform deviate U and accept Y if $Ucg_Y(Y) \leq f_X(Y)$ ■

Question 3

- What is detailed balance?
 1. It means you visit all possible values of the random variable equally often
 2. It is a Markov Chain that converges to a particular distribution
 3. It is the condition that the transition matrix is stochastic
 4. It is the equation $M_{ij}\pi_j = M_{ji}\pi_i$ ■
- Answer: 4. It is the equation $M_{ij}\pi_j = M_{ji}\pi_i$ ■

Question 4

- What statement is **false** about the transition matrix M_{ij} ?
 1. Each column sums to one, i.e. it is a stochastic matrix
 2. The absolute size of the eigenvalues are all less than or equal to 1
 3. Both its rows and columns sum to one, i.e. it is doubly stochastic
 4. The equilibrium state correspond to the eigenvectors with eigenvalue 1■
- Answer: 3. Both its rows and columns sum to one, i.e. it is doubly stochastic—**this is not true**■

Question 5

- What is the difference between the Metropolis algorithm and Metropolis-Hastings
 1. They are the same algorithm
 2. In Metropolis-Hastings you chose the proposal move from a non-symmetric distribution which you compensate for in the acceptance criteria
 3. You choose a transition probability that does not satisfy detailed balance
 4. You perform a molecular dynamics simulation to obtain the proposal move■
- Answer: 2.In Metropolis-Hastings you chose the proposal move from a non-symmetric distribution which you compensate for in the acceptance criteria ■

Question 6

- What statement is *false* about MCMC
 1. Once you have reached an equilibrium state every state after that is an independent deviate
 2. It is useful for Bayesian inference because you don't have to compute the marginal likelihood
 3. Generally it is difficult to determine if you have reached an equilibrium state
 4. To compute expectation you can use every state once you have reached equilibrium, but you need to count states if they are accepted or rejected, as well as run for a long time■
- Answer: 1. Once you have reached an equilibrium state every state after that is an independent deviate—this is **false**■